

## Chips (chips)

William is crazy about cosmic poker! In this game, coins of three colors ( $r, g, b$ ) are used instead of cards. The game is simple, the casino defines two numbers  $A$  and  $C$  as coefficients for calculating bets, and the player's win is calculated with the formula:

$$A \cdot (r^2 + g^2 + b^2) + C \cdot \min \{r, g, b\}$$

where  $r, g, b$  are the number of red, green and blue chips in the game.



Figure 1: A bunch of cosmic poker chips.

William is facing the following task: there are already  $r$  red,  $g$  green and  $b$  blue chips on the table, but with **one last move**, he can add on the table **exactly one chip** of any color. Help him choose a colour to maximize his win.

Moreover, William wants to solve this problem for  $T$  different game configurations.

Among the attachments of this task you may find a template file `chips.*` with a sample incomplete implementation.

### Input

The first line contains the only integer  $T$ , the number of games to solve. Each of the following  $T$  lines represents one game configuration and is formed by 5 integers:  $A$ ,  $C$ ,  $r$ ,  $g$  and  $b$ .

### Output

You need to write  $T$  lines, each of them should contain the color that William should choose for the corresponding game; in other words, each line should be one of the following string: RED, GREEN or BLUE.

## Constraints

- $1 \leq T \leq 10\,000$ .
- $1 \leq A, C \leq 10$ .
- $0 \leq r, g, b \leq 15$ .
- In case of many optimal solutions, print any of them.

## Scoring

Your program will be tested against several test cases. Your score is proportional to the number of correctly solved test cases.

## Examples

| input                                     | output               |
|-------------------------------------------|----------------------|
| 3<br>2 10 2 4 4<br>1 2 3 4 5<br>4 2 7 7 7 | RED<br>BLUE<br>GREEN |

## Explanation

In **the first scenario** it's better to add a red chip, the calculated winning is 112.